

Supplementary Material

A national inventory of groundwater well and lithology data for the
conterminous United States

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S1 Harvest-mode taxonomy

State subsurface data hides behind about ten distinct access modes (Table S1). The method is to identify which mode a state uses, then apply the matching harvester. Cost decreases up the digital end.

Table S1: The ten harvest modes encountered, with example states.

#	Mode	How	Cost	States
1	Bulk REST pagination	resultOffset paging → FlatGeobuf	free	most ArcGIS layers
2	objectIds-batch fallback	returnIdsOnly then objectIds batches	free	MS (181k recovered)
3	Bulk file download	CSV / GeoPackage / shape-file	free	CA, MN, MO/MSDIS
4	Inline structured lithology	interval table or wide cols in record	free	MT, AL, MO, KY, IL-bulk
5	HTML-table scrape	server-rendered lithology table per well	free	OK, IA
6	Constructible text-layer PDF	born-digital per-well PDF → pdftotext	~free	IL
7	Per-well scanned PDF (URL in record)	direct PDF/image URL → vision model	\$	OR, WI, FL, NV, SD, KS
8	Detail-page hop → scanned doc	record → page → scan	\$ + recon	ID, NY, UT, VT, NH, TN
9	Per-county framework service	aquifer-study framework, not logs	n/a	CO CGS (4 counties)
10	Other modality / by-request	geophysical logs; email/FOIA	n/a	CO DWR; NJ, MA, ND, ...

S2 Per-state attempt record

Table S2 is the engineering record of what was tried per state. Status: ✓ harvested; ○ partial/slice; ▷ calibrated, not fully mined; × blocked/remaining; — out of scope.

Table S2: Per-state harvest-attempt record.

State	Source	Mode	Status	Notes
MI	Wellogig (EGLE)	4	✓	5.47M intervals; canonical schema
TX	TWDB SDR	4	✓	5.29M intervals; bulk lithology/strata files
MN	CWI C5ST	3	✓	2.84M intervals (GeoPackage)
IL	ISGS → wa- ter_summary	4→6	✓	2.20M intervals; text-layer PDF (free)
MT	GWIC	4	✓	1.54M intervals, 100% non-null
CA	OSWCR (4-CSV)	3	✓	1.33M intervals; section-center geometry
WI	DNR Re- portViewer	6/7	✓	1.26M intervals; born-digital triage
OK	OWRB printre- port	5	✓	1.10M intervals; HTML scrape
MO	DNR / MSDIS	3/4	✓	947k intervals
MA	MassDEP	6	✓	742k intervals; born-digital report PDFs
VT	ANR	3	✓	377k intervals; structured grid
TN	TDEC	5	✓	309k intervals; viewer-resolved
KY	KGS	4	✓	168k intervals; inline
OH	ODNR	5	✓	162k intervals; viewer-resolved
IA	GeoSam	5	✓	118k intervals (HTML scrape)
NM	OSE	7	○	52k intervals; structured API, WAF- limited (partial)
AL	NGWMN	4	○	41k intervals, sparse source
OR	OWRD vault	7	▷	VLM route; characterized by sampling, not bulk-harvested
FL	SJRWMD DO- CLINK	7	▷	VLM route; sampled; one WMD only
NV	NDWR images	7	▷	VLM route; sampled
SD	DANR images	7	▷	VLM route; sampled
KS	KGS Azure blob	7	▷	VLM route; sampled
ID	IDWR Laserfiche	8	▷	VLM route; sampled
NY	DEC WCR	8	×	scanned; resolver not wired
UT	waterrights wl- browse	8	×	~1.1k geologic-log wells only
CO	CGS / DWR	9/10	×	geophysical + framework, weak driller- lithology
WA	Ecology	7	×	1.18M wells; report-PDF base URL needed
MS	state	2	×	points harvested; lithology not yet wired
NE, AZ, WY, NH, ME, VA, SC	various	7/8	×	points harvested; lithology resolvers not wired
NJ, ND, MD, RI, WV, NC, GA, AR, LA	—	10	×	by-request / app-gated
AK, HI	—	—	—	out of scope (CONUS only)

S3 Hydraulic-ingredient coverage

Measured conductivity is never in a driller’s log; it must be derived from a pump test (specific capacity $\rightarrow$ transmissivity) and a screened thickness. Table S3 audits, per ingested state, which raw ingredients exist and how populated they are (cell = max non-null % in a 1,500-row sample of any matching field; name-based lower bound). Static water level and total depth form a broad backbone; the pump-test fields are scarce.

Table S3: Hydraulic-ingredient coverage (selected states; “–” = no matching field).

State	SWL	yield	drawdown	spec. cap.	transmiss.	depth
WI	1%	1%	1%	1%	–	3%
NV	61%	9%	5%	5%	5%	97%
KS	100%	100%	–	–	–	100%
MO	91%	81%	–	–	–	100%
OR	100%	72%	–	–	–	96%
AZ	87%	67%	67%	–	–	91%
ID	89%	63%	–	–	–	74%
VA	100%	100%	–	–	–	–

Reading it: direct specific capacity appears in only NV and WI; drawdown in AZ, NV, WI; measured transmissivity in NV alone. A national derived-conductivity layer is therefore feasible only where yield and drawdown co-occur; elsewhere conductivity must fall back to lithology-facies estimates anchored to the observed log. Full matrix and generator: `hydraulic_coverage.md` and `audit_hydraulic_coverage.py` (under `national_gw_inventory/`).

S4 Extraction schema and validation

The forced JSON schema returns, per log: ordered intervals (top/bottom feet, verbatim description, normalized class), total depth, static water level, a legibility flag, and a self-confidence score. Validation across the seven calibrated states found faithful transcription of driller shorthand (including preserved misspellings), legibility flags on illegible scans, and *empty* arrays rather than fabricated intervals on blank logs — the key trust signal. A quantitative ground-truth comparison against a state agency’s own hand-transcriptions ($n = 1,499$ wells) is reported in the main text.

S5 Data sources and access provenance

This section documents, for every state we touched, exactly how its well and lithology data were accessed: the custodian agency and dataset, the access endpoint (a bulk-file portal or, more often, an ArcGIS/REST feature service or a per-well report endpoint), the access method, the harvested point count, the route by which lithology is recovered, and the date the endpoint was last accessed and verified (2026-06-24 throughout). Table S4 is the citable per-state ledger; each agency portal also carries a data-citation entry in the bibliography (citekeys `data_<st>_<agency>`).

The lithology route falls into three provenance *tiers*, ordered by extraction cost. **(a) Structured / bulk** (no vision model): lithology arrives already machine-readable — a bulk file (CA OSWCR CSVs, MN CWI GeoPackage, MO/MSDIS shapefile), an inline interval table or wide columns in the REST record (MI, MT, AL, KY, IL-bulk), or a server-rendered HTML grid scraped per well (OK, IA, VT, NM). **(b) Born-digital text-layer PDF** (cheap text-only LLM): the per-well report is a digital PDF whose strata table survives `pdftotext`, so no image tokens are needed (TX, OH, NY, IL via the `water_summary` endpoint, MA). **(c) Scanned PDF** (vision-language model): the log is a raster scan of a paper form and must be read with a vision model (WA, MS, NE, UT, ID, OR, WI, FL, NV, SD, KS). For ten states the data is present but was, at audit time, *out of reach* of an automated anonymous client, and we record *why*, because the obstruction is procedural rather than an absence of data: TN, NH, and AZ return HTTP 403 to our primary datacenter IP (an AWS ELB/WAF, an Akamai edge rule, and a Cloudflare JS/Turnstile challenge, respectively). These are IP-reputation blocks, not data restrictions, and are only partially circumventable: re-issuing the requests from a fresh AWS EC2 elastic IP cleared *TN* — whose report page in fact renders a *structured HTML* lithology table requiring no vision model, from which we then harvested 77,836 wells — whereas the Akamai (NH) and Cloudflare (AZ) rules fingerprint datacenter IP ranges generally and still returned 403, so those two require a residential or headless-browser egress. WY places its statements of completion behind an eGov login with no anonymous handler; PA’s scanned records exist only as GIF images keyed by an internal identifier with no public record-to-image mapping; and VA, ME, and SC publish only located-well tables, with the actual completion reports held in internal storage or released by request. None of these is a confidentiality restriction — the records are public; they are simply not yet harvestable by an automated, anonymous client.

Table S4: Per-state groundwater data sources, access methods, and provenance. “Wells” is the harvested point count; “–” marks states for which only locations (not logs) are reachable. Method vocabulary: bulk file; REST/ArcGIS; structured HTML; text PDF (born-digital, text layer); scanned PDF (VLM); blocked (WAF/login); by-request gap. All endpoints last accessed 2026-06-24.

State	Agency / dataset	Access endpoint	Method	Wells	Lithology route	Last acc.
MI	Wellogic / EGLE (Michigan Department of Environment, Great Lakes, and Energy (EGLE), 2026)	state DB export	bulk file	5.77M [†]	inline interval table (a)	2026-06-24

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Table S4 (continued)

State	Agency / dataset	Access endpoint	Method	Wells	Lithology route	Last acc.
MN	MN Health / CWI (Minnesota Department of Health and Minnesota Geological Survey, 2026)	CWI C5ST GeoPackage	bulk file	2.84M [†]	bulk strata table (a)	2026-06-24
MT	MBMG GWIC (Montana Bureau of Mines and Geology, 2026)	mbmgmap.mtech.edu/.../GWIC_Database/FeatureServer/13	REST/ArcGIS	30,278	inline interval table (a)	2026-06-24
CA	CA DWR OS-WCR (California Department of Water Resources, 2026)	data.cnra.ca.gov/dataset/well-completion-reports	bulk file	1.11M	4-CSV union (a)	2026-06-24
MO	MO DNR / MS-DIS (Missouri Department of Natural Resources, Missouri Geological Survey, 2026)	gis.dnr.mo.gov/.../water_wells/MapServer/1	REST + bulk	194,607	wide-column un-pivot (a)	2026-06-24
IL	ISGS ILWATER (Illinois State Geological Survey, 2026)	maps.isgs.illinois.edu/.../ILWATER/.../MapServer/2	REST + text PDF	390,443	bulk col. (a) + text PDF (b)	2026-06-24
KY	KGS (Kentucky Geological Survey, 2026)	kgs.uky.edu/.../KYWaterWells/MapServer/1	REST/ArcGIS	505,714	inline columns (a)	2026-06-24
AL	GSA NGWMN (Geological Survey of Alabama, 2026)	map.ogb.state.al.us/.../NGWMN/Construction/MapServer/0	REST/ArcGIS	38,408	inline interval table (a)	2026-06-24
IA	Iowa GS GeoSam (Iowa Geological Survey (IHR), University of Iowa, 2026)	services3.arcgis.com/.../GeoSam/FeatureServer/0	structured HTML	99,729	HTML stratigraphy scrape (a)	2026-06-24
OK	OWRB (Oklahoma Water Resources Board, 2026)	owrb.csa.ou.edu/.../Groundwater_Wells/MapServer/4	structured HTML	243,801	printreport.html scrape (a)	2026-06-24

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Table S4 (continued)

State	Agency / dataset	Access endpoint	Method	Wells	Lithology route	Last acc.
VT	VT DEC anr-web (Vermont Department of Environmental Conservation, 2026)	anrweb.vt.gov/DEC/IWIS/	structured HTML	120,051	lithology grid scrape (a)	2026-06-24
NM	NM OSE NMWRRS (New Mexico Office of the State Engineer, 2026)	services2.arcgis.com/.../OSE_Points_of_Diversion/.../0	structured HTML	278,408	POD-report strata grid (a)	2026-06-24
TX	TWDB SDR (Texas Water Development Board, 2026)	www3.twdb.texas.gov/apps/.../GetReports.aspx	text PDF	680,660	SSRS Well Report PDF (b)	2026-06-24
OH	ODNR (Ohio Department of Natural Resources, Division of Soil and Water Resources, 2026)	gis2.ohiodnr.gov/.../waterwells/MapServer/0	text PDF	478,224	born-digital log PDF (b)	2026-06-24
NY	NYSDEC (New York State Department of Environmental Conservation, 2026)	services6.arcgis.com/.../Water_Wells/FeatureServer/28	text PDF	144,498	RPTURL → Completion.pdf (b)	2026-06-24
MA	MassDEP EEA (Massachusetts Department of Environmental Protection (EEA), 2026)	eeaonline.eea.state.ma.us/.../WellDrilling/	text PDF	214,806	born-digital report PDF (b)	2026-06-24
WA	WA Ecology (Washington State Department of Ecology, 2026)	<a href="http://appswr.ecology.wa.gov/wellconstruction/WellReport/<ID>">appswr.ecology.wa.gov/wellconstruction/WellReport/<ID>	scanned PDF	1.18M	embedded scan → VLM (c)	2026-06-24
MS	MDEQ (Mississippi Department of Environmental Quality, 2026)	landandwater.deq.ms.gov/DrillScans/	scanned PDF	181,279	DrillScans archive → VLM (c)	2026-06-24

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State	Agency / dataset	Access endpoint	Method	Wells	Lithology route	Last acc.
NE	NE DNR/DWEE (Nebraska Department of Natural Resources, 2026)	nednr.nebraska.gov/dynamic/PublicScans/	scanned PDF	266,555	PublicScans 2-hop → VLM (c)	2026-06-24
UT	UT DWRI (Utah Division of Water Rights, 2026)	waterrights.utah.gov/cgi-bin/docview.exe?Folder=welllog%3CWIN%3E	scanned PDF	94,258	TIFF pages → PDF → VLM (c)	2026-06-24
ID	IDWR (Idaho Department of Water Resources, 2026)	gis.idwr.idaho.gov/.../Groundwater/Wells/.../0 (Laserfiche)	scanned PDF	207,787	detail-hop → scan → VLM (c)	2026-06-24
OR	OWRD (Oregon Water Resources Department, 2026)	gis.wrd.state.or.us/.../Wells_by_Theme_WGS84/.../0	scanned PDF	594,185	well_log_urls → VLM (c)	2026-06-24
WI	WI DNR (Wisconsin Department of Natural Resources, 2026)	services5.arcgis.com/.../GW_SAMPLE_PT_EXT_VAR_gdb2/.../0	scanned PDF	871,268	WELL_CONSTRUCTURL → VLM (c)	2026-06-24
FL	SJRWMD (Florida St. Johns River Water Management District, 2026)	services.arcgis.com/s8wtJX9suxFen6TA/.../0	scanned PDF	410,841	DOCLINK → VLM (c)	2026-06-24
NV	NDWR (Nevada Division of Water Resources, 2026)	arcgis.water.nv.gov/.../Well_Driller_Reports/.../0	scanned PDF	144,746	WellLogReports → VLM (c)	2026-06-24
SD	SD DANR (South Dakota Department of Agriculture and Natural Resources, 2026)	arcgis.sd.gov/.../DENR/WellLogs_Public/MapServer/0	scanned PDF	78,246	LINK → VLM (c)	2026-06-24
KS	KGS WWC5 (Kansas Geological Survey, 2026)	services2.arcgis.com/.../Water_Wells_WWC5/.../0	scanned PDF	313,546	FILE_URL → VLM (c)	2026-06-24
AZ	ADWR Wells55 (Arizona Department of Water Resources, 2026)	infoshare.azwater.gov (DocuShare)	blocked (WAF)	239,567	Cloudflare 403; needs egress	2026-06-24

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Table S4 (continued)

State	Agency / dataset	Access endpoint	Method	Wells	Lithology route	Last acc.
TN	TDEC DWR (Tennessee Department of Environment and Conservation, 2026)	<code>dataviewers.tdec.tn.gov</code> (APEX)	structured HTML	77,836	fresh AWS egress cleared ELB/WAF	2026-06-24
NH	NHDES On-eStop (New Hampshire Department of Environmental Services, 2026)	<code>www4.des.state.nh.us/DESOnestop/WLSDetail.aspx</code>	blocked (WAF)	111,476	Akamai 403; needs egress	2026-06-24
WY	WY SEO e-Permit (Wyoming State Engineer’s Office, 2026)	<code>seoweb.wyo.gov/e-Permit/</code>	blocked (login)	208,813	login-gated completion scans	2026-06-24
PA	PA DCNR PaGWIS (Pennsylvania Department of Conservation and Natural Resources, Bureau of Geological Survey, 2026)	<code>iframeapps.dcnr.state.pa.us/.../WellReports/</code>	by-request gap	417,259	GIF, no record→image map	2026-06-24
VA	VA Energy DGMR (Virginia Department of Energy, Division of Geology and Mineral Resources, 2026)	<code>energy.virginia.gov/.../DGMR/VA_Water_Wells/.../0</code>	by-request gap	19,937	reports by request (internal)	2026-06-24
ME	Maine GS (Maine Geological Survey (DACF), 2026)	<code>services1.arcgis.com/.../MGS_Wells_Database/.../0</code>	by-request gap	94,685	scanned cards by request	2026-06-24
SC	SC DES/DNR (South Carolina Department of Environmental Services / Department of Natural Resources, 2026)	<code>services.arcgis.com/.../SCDNR_Coastal_Plain_Well_Inventory/0</code>	by-request gap	14,146	logs by request (photo-copy)	2026-06-24

[†] MI and MN counts are harmonized lithology *intervals* (bulk state exports predate the REST point harvest); all other “Wells” values are harvested point counts.

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